

Technical Document #972: Database Systems - Comprehensive Overview

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Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Database caching stores frequently accessed data in memory. Redis and Memcached are popular caching solutions.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Database indexing improves query performance by creating data structures that allow faster data retrieval.
- Database replication copies data across multiple servers.
- Sharding distributes data across multiple database servers.